ETALE MORPHISMS LIFT UNIQUELY TO UNIVERSAL FIRST ORDER THICKENINGS AND REFLECT THE UNIVERSAL PROPERTY

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3024-3041 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 3043-3069, the complete author proof (25 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Universal first order thickenings, etale morphisms of algebraic spaces, and topological invariance of the etale site; the construction is retained.

Source reference. Original lemma equation-universal-first-order-thickening: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2944–2952. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-check-universal-first-order-thickening: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2964–2976. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-representable-property-formally-property: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2529–2537. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma theorem-topological-invariance: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1316–1326. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $Z \to Y \to X$ be morphisms of algebraic spaces over S. Assume $Z \to Y$ is étale.*

(1) *If $Y \subset Y'$ is a universal first order thickening of Y over X, then the unique étale morphism $Z' \to Y'$ such that $Z = Y \times_{Y'} Z'$ (see Theorem theorem-topological-invariance) is a universal first order thickening of Z over X.*

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

(2) *If $Z \to Y$ is surjective and $(Z \subset Z') \to (Y \subset Y')$ is an étale morphism of first order thickenings over X and Z' is a universal first order thickening of Z over X, then Y' is a universal first order thickening of Y over X.*

Proof. Proof of (1). By Lemma lemma-check-universal-first-order-thickening it suffices to consider solid commutative diagrams (equation-universal-first-order-thickening) with T' an affine scheme. The composition $T \to Z \to Y$ lifts uniquely to $T' \to Y'$ as Y' is the universal first order thickening. Then the fact that $Z' \to Y'$ is étale implies (see Lemma lemma-representable-property-formally-property) that $T' \to Y'$ lifts to the desired morphism $a' : T' \to Z'$.

Proof of (2). Let $T \subset T'$ be a first order thickening over X and let $a : T \to Y$ be a morphism. Set $W = T \times_Y Z$ and denote $c : W \to Z$ the projection Let $W' \to T'$ be the unique étale morphism such that $W = T \times_{T'} W'$, see Theorem theorem-topological-invariance. Note that $W' \to T'$ is surjective as $Z \to Y$ is surjective. By assumption we obtain a unique morphism $c' : W' \to Z'$ over X restricting to c on W. By uniqueness the two restrictions of c' to $W' \times_{T'} W'$ are equal (as the two restrictions of c to $W \times_T W$ are equal). Hence c' descends to a unique morphism $a' : T' \to Y'$ and we win. □